

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM

August 2012

ALGEBRA (Ph.D. Version)

Instructions to the student

- Answer all six questions; each will be assigned a grade from 0 to 10.
 - Use a different booklet for each question. Write the problem number and your **code number** (not your name) on the outside of the booklet.
 - Keep scratch work on separate pages in the same booklet.
-

1. Let G be a finite group. Recall that G is solvable if there are subgroups

$$1 \subseteq G_1 \subseteq G_2 \subseteq \cdots \subseteq G_s = G$$

with G_i normal in G_{i+1} and G_{i+1}/G_i abelian for $1 \leq i \leq s-1$. It is a standard fact that a subgroup of a solvable group is solvable. Another standard fact that might be useful is that if N is a normal subgroup of G and H is a characteristic subgroup of N (that is, H is mapped to itself by all automorphisms of N), then H is normal in G .

(a) Let N be a normal subgroup of G . Let N' be the subgroup of N generated by elements of the form $aba^{-1}b^{-1}$ with $a, b \in N$ (so N' is the commutator subgroup of N). Show that N' is a normal subgroup of G .

(b) Suppose that G is solvable and N is a minimal normal subgroup of G (this means that $N \neq 1$ and if M is a normal subgroup of G with $1 \subseteq M \subseteq N$, then $M = 1$ or $M = N$). Show that N is abelian.

(c) Suppose that G is solvable and N is a minimal normal subgroup of G . Show that

$$N \simeq (\mathbf{Z}/p\mathbf{Z})^r$$

for some prime p and some $r \geq 1$.

2. Let T be a linear transformation of a finite-dimensional complex vector space V . Recall that we define $\text{rank}(T) = \dim(\text{Image}(T))$. Let I be the identity transformation of V .

(a) Show that $\text{rank}(T) + \text{rank}(I - T) \geq \dim V$.

(b) Show that if $\text{rank}(T) + \text{rank}(I - T) = \dim V$ then $T^2 = T$.

3. Let R be a unique factorization domain.

(a) Let $\pi \in R$ be irreducible. Show that πR is a prime ideal.

(b) Show that every nonzero prime ideal of R contains a prime ideal of the form πR for some irreducible π .

(c) Suppose every non-zero prime ideal of R is maximal. Let $S = \{r_i\}_{i \in I}$ be a set of elements of R . Show that if the ideal generated by S is not R , then there is an irreducible π such that π divides r_i for all i .

(d) Suppose every non-zero prime ideal of R is maximal. Show that R is a principal ideal domain.

4. Let P_1 and P_2 be projective modules (over a ring R). Suppose that there are exact sequences

$$0 \longrightarrow K_1 \xrightarrow{f_1} P_1 \xrightarrow{g_1} M \longrightarrow 0 \quad \text{and} \quad 0 \longrightarrow K_2 \xrightarrow{f_2} P_2 \xrightarrow{g_2} M \longrightarrow 0.$$
 It can be

shown (that is, don't prove this) that there are maps $\alpha : K_1 \rightarrow K_2$ and $\beta : P_1 \rightarrow P_2$ such that the following diagram commutes:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & K_1 & \xrightarrow{f_1} & P_1 & \xrightarrow{g_1} & M & \longrightarrow & 0 \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow = & & \\ 0 & \longrightarrow & K_2 & \xrightarrow{f_2} & P_2 & \xrightarrow{g_2} & M & \longrightarrow & 0 \end{array}$$

(a) Let $F : K_1 \rightarrow P_1 \oplus K_2$ be defined by $F(x) = (f_1(x), \alpha(x))$ and $G : P_1 \oplus K_2 \rightarrow P_2$ be defined by $G(p, k) = \beta(p) - f_2(k)$. Show that the sequence of modules

$$0 \longrightarrow K_1 \xrightarrow{F} P_1 \oplus K_2 \xrightarrow{G} P_2 \longrightarrow 0$$

is exact.

(b) Show that $K_1 \oplus P_2 \simeq K_2 \oplus P_1$.

(c) Show that if A and B are R -modules such that $A \oplus B$ is finitely generated, then A is finitely generated.

(d) Suppose that I is an ideal of R that is not finitely generated (as an R -module). Let $M = R/I$. Show that there is no sequence $0 \rightarrow K \rightarrow P \rightarrow R/I \rightarrow 0$ with both K and P finitely generated and P projective.

5. Let $G = \{\pm 1, \pm i, \pm j, \pm k\}$ be the quaternion group (so $i^2 = j^2 = k^2 = -1$ and $ij = -ji = k$). Suppose $L/\mathbb{Q}$ is a Galois extension with $\text{Gal}(L/\mathbb{Q}) \simeq G$.

(a) Show that there are 3 subextensions $F/\mathbb{Q}$ such that $[F : \mathbb{Q}] = 2$.

(b) Regard L as a subfield of $\mathbb{C}$. Let c denote complex conjugation. Show that there exists $\sigma \in G$ such that $\sigma^2 = c$.

(c) Show that c acts trivially on the quadratic subfields of L (whose existence was proved in part (a)). (*Hint:* Look at the restriction of σ to these subfields.)

6. Let $\rho : G \rightarrow \text{GL}_2(\mathbb{C})$ be a two-dimensional complex representation of the finite group G . Let V be the 4-dimensional vector space of 2×2 complex matrices, and let G act on V by

$$\tilde{\rho}(g)(M) = \rho(g)M\rho(g)^{-1}$$

for $M \in V$.

(a) Show that if ρ is irreducible then $\tilde{\rho}$ contains the trivial representation exactly once.

(b) Show that if ρ is the sum of two distinct one-dimensional representations, then $\tilde{\rho}$ contains the trivial representation exactly twice.

(c) Show that if ρ is the sum of two equal one-dimensional representations, then $\tilde{\rho}$ equals the sum of four copies of the trivial representation.